

Q-LULAN WATER SOLUBLE POWDER 300mg/g

Ingredient(s):

Each g contains :
Sulfachloropyrazine Sodium Monohydrate 342.146mg
(eq. Sulfachloropyrazine 300mg)

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

An anti-coccidiosis sulfa-drug which is widely used for livestock coccidiosis. The drug may influence synthesis of dihydrofolacin so that restrain the growth of bacteria and coccidium. It's action property similar with sulfaquinoxaline but have stronger anti-bacterial action.

Absorption of a sulfonamide and passage from its site of administration to blood are governed by passive diffusion. Most sulfonamides are rapidly absorbed from the GI tract. Effects of age, dosage form, and environment on absorption of sulfonamides following oral administration have been characterized.

Once sulfonamides have been absorbed, they undergo rapid mixing in the bloodstream. Distribution of sulfonamides into various tissues and fluids occurs as a result of passive diffusion. Protein-bound sulfonamides are not bacteriostatic and do not pass from the blood to other extracellular fluids. However, the unbound drug fraction is freely diffusible. The ionization state of a sulfonamide has a profound influence upon further distribution once the drug has crossed the capillary endothelium. Unbound, non-ionized drugs cross biologic membranes more rapidly than their bound or ionized forms. Eventually, a dynamic equilibrium is established between the sulfonamide in blood, other extracellular fluids, and intracellular fluids. The rate at which such an equilibrium is established is dependent upon the blood supply to specific tissues. Concentrations of sulfonamides in tissues are consistently lower than those observed in plasma. The blood-brain barrier is permeable to the non-ionized fraction of freely diffusible sulfonamides. As a result, concentration of these compounds in cerebrospinal fluid is equivalent to concentration of unbound and non-ionized drugs in blood. Sulfonamides passively diffuse into milk from blood following oral administration.

Sulfonamides are extensively metabolized in the animal body. Acetylation, deacetylation, oxidation, conjugation with sulfate or glucuronic acid, deamination, and cleavage of their heterocyclic rings have been reported. In general, acetylated sulfonamides are the major urinary excretion product following enteral or parenteral administration of sulfonamides to cattle, sheep, and swine. Oxidation of the benzene and heterocyclic rings of sulfonamides is an important metabolic pathway in sheep and cattle. Sulfonamides and their metabolites are often conjugated with glucuronic acid and sulfates. Conjugation often occurs at sites of previous hydroxylation (benzene ring, heterocyclic ring, or aliphatic constituents of heterocyclic rings) or at the N1 or N4 positions. Ring cleavage appears to be a metabolic pathway of minor importance in most animals, as only small amounts of cleavage products are excreted in urine. In general, acetylated, hydroxylated, and conjugated forms of the sulfonamides exhibit marked decrease in antibacterial activity when compared to their parent compounds. Acetylation reduces solubility of a sulfonamide and increases the chance for renal damage. Usually, hydroxylated and conjugated forms are less likely to precipitate in urine but represent a pathway where an antibacterially active drug can be removed more rapidly from blood and tissues and excreted in urine.

Sulfonamides are eliminated in urine, feces, bile, milk, sweat, and tears. However, the kidney is primarily involved in excretion of these drugs. The rate of sulfonamide excretion is primarily determined by the net consequences of glomerular filtration, renal tubular resorption, and active tubular secretion. All sulfonamides existing in an ultrafiltrable state in plasma are filtered from the blood at the renal glomerulus. Rate of filtration is proportional to concentration of unbound drug and metabolites in blood. The renal tubular epithelium is a lipid boundary that permits passive diffusion of non-ionized sulfonamides or sulfonamide metabolites from the tubular lumen to the peritubular fluids and eventually to blood. The amount of a sulfonamide available for resorption (non-ionized form) by passive diffusion is dependent on the pH of fluids within the tubular lumen. Increases persistence of the drug in the body is attributed to renal processes of resorption that recycle filtered drug back into blood.

Indication(s):

For treatment of coccidiosis in poultry due to infection with *Eimeria* species, fowl typhoid due to infection with *Salmonella gallinarum* and fowl cholera due to infection with *Pasteurella multocida*.

Target Species

Chicken and turkeys.

Dosage and Administration:

General: Dissolve 1g into every L of drinking water, or apply 170mg to every kg body weight. Treat for 3 consecutive days and repeat after a 2 days break if necessary.

Coccidiosis in chicken caused by *E. tenella* and *E. necatrix*: 1.5~2.0g/L of drinking water.

Treatment of cholera and typhoid: 1~2g/ L of drinking water for at least 5 days.

Orally via drinking water.

Contraindications:

- 1) Fluid therapy for treatment of renal damage produced by sulfonamides is contraindicated in anuric animals because of the attendant risk of overhydration.
- 2) Except under special circumstances, sulfonamides should not be used in animal that have had a previous reaction to sulfonamides because they may be much more likely to react on subsequent administration.
- 3) Risk-benefit should be considered in hepatic function impairment patients since systemically absorbed sulfonamides are metabolized by the liver; delayed biotransformation may increase the risk of adverse effects.
- 4) Risk-benefit should be considered in renal function impairment patients since systemically absorbed sulfonamides are renally excreted; delayed elimination could cause accumulation of sulfonamide and metabolites, increasing the risk of adverse effects.

Precaution(s) / Warning(s):

- 1) For use in drinking water of poultry. Prepare fresh water daily.
- 2) Avoid underdosing. Sick birds may drink less water than normal. When such birds can be isolated the proportion of Q-lulan in the water can be proportionally increased.
- 3) Avoid overdosing. At high ambient temperatures birds drink more water than usual, and thus the concentration of Q-lulan in the water should be decreased.

Interaction with Other Medicaments

- 1) Concurrent use of anticoagulants (coumarin- or indandione-derivative), anticonvulsants (hydantoin), and antidiabetic agents may be displaced from protein binding sites and/or their metabolism may be inhibited by some sulfonamides, resulting in increases or prolonged effects and/or toxicity.
- 2) Concurrent use of bone marrow depressants with sulfonamides may increase the leukopenic and/or thrombocytopenic effects.
- 3) Concurrent use of cyclosporine with sulfonamides may increase the metabolism of cyclosporine, resulting in decreased plasma concentrations and potential transplant rejection, and additive nephrotoxicity.
- 4) Concurrent use of hemolytics with sulfonamides may increase the potential for toxic side effects.
- 5) Concurrent use of hepatotoxic medicines with sulfonamides may result in an increased incidence of hepatotoxicity.
- 6) In acid urine, methenamine breaks down into formaldehyde, which may form an insoluble precipitate with certain sulfonamides, especially those that are less soluble in urine, and may also increase the danger of crystalluria.
- 7) The effects of methotrexate may be potentiated during concurrent use with sulfonamides because of displacement from plasma protein binding sites.
- 8) Phenylbutazone may displace sulfonamides from plasma protein binding sites, increasing sulfonamide concentrations.
- 9) Since bacteriostatic drugs (like sulfonamides) may interfere with the bactericidal effect of penicillins in the treatment of meningitis or in other situations where a rapid bactericidal effect is necessary, it is best to avoid concurrent therapy.

Pregnancy and Lactation

- 1) Sulfonamides cross the placenta in pregnant animals. Some teratogenic effects have been seen when very high doses were given to pregnant mice and rats.
- 2) Sulfonamides are distributed into milk; however, the sulfonamides that are clinically relevant to food-producing animals are distributed into milk in concentrations too low to be therapeutic but high enough to produce residues. For lactating dairy cattle, concentration of the active parent compound of sulfamethazine, measured at a specific time in milk, is about 20% of the concentration in the blood.

Side Effect(s) / Adverse Reaction(s):

- 1) Disturbances of the kidney and urinary tract with syndromes of crystalluria, hematuria, and obstruction of renal tubules are the commonest sequelae. Rapidly excreted sulfonamides are more likely to produce crystalluria than those excreted more slowly, but the intrinsic solubility of individual sulfonamides and their metabolites influences crystallization. Renal damage associated with use of sulfonamides having low intrinsic urinary solubility can be reduced by using a mixture of two or three of these compounds.
- 2) Anorexia, depression, hematuria, crystalluria, renal colic, frequent attempts to urinate, and elevation of blood urea nitrogen may be observed following renal precipitation of sulfonamides.
- 3) Other adverse reactions to sulfonamides in dogs include keratoconjunctivitis sicca, bone marrow suppression, and cutaneous allergic reactions.

Environmental Property

None has been mentioned.

Symptoms and Treatment for Overdosage, and Antidote(s):

Toxicities secondary to acute overdose of sulfonamides are not typically reported. Side effects may be more likely to occur with high doses and long-term administration, but are seen at recommended doses as well. Recommended treatment consists of parenteral epinephrine, oxygen administration and respiratory support.

Shelf-Life:

4 years.

Storage Condition(s):

Store at temperature below 30°C.

Product Description and Packing(s):

A white to off-white color powder.

Al. Pouch of 180g

For food producing animals product:

Chicken: 14 days before slaughter.

Turkeys: 21 days before slaughter.

Eggs: 11 days before human consumption.

Maximum Residual Limit (MRL):

Nil



Y. S. P.

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